

Q3 – Prepare a Basic Cloud Workload Fitment Decision Matrix

Scenario:

A company has four workloads: an internal HR system, a public product website, a legacy desktop application and a data backup requirement. Management wants to know which workloads are suitable for cloud migration first. You must prepare a simple cloud readiness and workload fitment decision matrix.

Problem Statement:

1. Identify cloud-ready and non-cloud-ready workloads.
2. Apply basic factors such as scalability, security, dependency and cost.
3. Rank workloads based on migration priority.
4. Recommend which workload should move to cloud first.

Scenario

A company wants to evaluate which of its existing workloads should be migrated to the cloud first. The decision is based on four factors:

- **Scalability** – Ability to handle changing demand.
- **Security Requirement** – Sensitivity of data and access control needs.
- **Dependency** – Reliance on legacy systems or local infrastructure.
- **Cost Benefit** – Potential cost savings after migration.

PS-1. Identify cloud-ready and non-cloud-ready workloads.

Workload	Scalability	Security	Dependency	Cost Benefit	Cloud Readiness	Migration Priority
Internal HR System	Medium	High	Medium	Medium	Moderate	Medium
Public Product Website	High	Medium	Low	High	Excellent	High
Legacy Desktop Application	Low	Medium	High	Low	Poor	Low
Data Backup Requirement	High	High	Low	High	Excellent	High

PS-2. Apply basic factors such as scalability, security, dependency and cost.

Scoring Scale:

- 1 = Low suitability
- 3 = Medium suitability
- 5 = High suitability

Workload	Scalability	Security	Dependency	Cost Benefit	Total Score	Cloud Readiness
Internal HR System	3	2	3	3	11	Moderate
Public Product Website	5	4	5	5	19	High
LegacyDesktop Application	2	3	1	2	8	Low
Data Backup Requirement	5	4	5	5	19	High

Justification:

1. Internal HR System:

- Stores employee records → requires stronger security and compliance.
- Moderate scalability requirements.
- May connect with internal databases.
- **Decision:** Suitable after planning or using a hybrid approach.

2. Public Product Website:

- Needs public accessibility and flexible scaling.
- Cloud supports high availability and traffic growth.
- Lower migration complexity.
- **Decision:** Best candidate for immediate migration.

3. Legacy Desktop Application

- Often tightly coupled to old infrastructure.
- May require redesign or modernization before cloud.
- Migration effort can be expensive.
- **Decision:** Keep for later.

4. Data Backup Requirement

- Cloud backup offers elastic storage and lower hardware maintenance.
- Easy to scale and cost-efficient.
- **Decision:** Migrate immediately.

PS-3. Rank workloads based on migration priority.

Migration Order

1. **Public Product Website** → Highest priority
2. **Data Backup Requirement** → Highest priority
3. **Internal HR System** → Medium priority
4. **Legacy Desktop Application** → Lowest priority

PS-4. Recommend which workload should move to cloud first.

1st Priority — Public Product Website

Recommendation: Move this workload to the cloud first.
Reason: It benefits the most from cloud capabilities such as scalability, public accessibility, high availability, and reduced infrastructure management. Migration effort is generally lower compared to other workloads.

2nd Priority — Data Backup Requirement

Recommendation: Migrate immediately after the website.
Reason: Cloud storage provides flexible capacity, improved reliability, lower maintenance effort, and cost-efficient backup management.

3rd Priority — Internal HR System

Recommendation: Migrate later with careful planning.

Reason: Since it handles sensitive employee information, stronger security controls and compliance checks are required before migration.

4th Priority — Legacy Desktop Application

Recommendation: Keep for the final migration phase.

Reason: High dependency on older infrastructure and possible compatibility limitations make direct cloud migration more complex.

Final Recommendation:

Public Product Website → Data Backup Requirement → Internal HR System → Legacy Desktop Application

Risk Note for Low-Priority Workload

Legacy Desktop Application – Risks

- Compatibility issues with cloud environments
- Higher migration and redesign effort
- Possible application downtime during transition
- Increased project cost if migrated without modernization

Risk Mitigation: Perform application assessment and modernization before cloud adoption.